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Research paper

Thermal analysis and design of solid energy storage systems using a modified lumped capacitance method

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HIGHLIGHTS

- We develop an unsteady model of a solid heat storage module using a modified lumped capacitance method.
- We derive an effective heat transfer coefficient for a cylindrical heat storage.
- We give an optimization design procedure for the solid storage module.
- We test a concrete storage module to evaluate the modeling work.
- An optimization design case is studied.

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ABSTRACT

Solid sensible heat storage is an attractive option for high-temperature storage applications in terms of the investment and maintenance costs. Typical solid thermal energy storage systems use a heat transfer fluid to deposit and extract heat as the fluid flows through a tubular heat exchanger embedded in the solid storage material. A one-dimensional unsteady model of the heat transfer and energy storage in a solid heat storage module was developed using a modified lumped capacitance method which is valid for large Biot numbers with the introduction of an effective heat transfer coefficient. Experiments using water as the heat transfer fluid and concrete as the storage material were used to validate the model. The model was used to analyze a cylindrical energy storage unit to study the energy storage in the storage module and optimize the design.

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1. Introduction

Thermal energy storage (TES) is indispensable for concentrating solar power (CSP) plant applications [1,2]. The main advantages of integrating a CSP system with thermal storage include extended utilization of the power block, improved dispatchability, and extended life expectancy of the components due to the reduction of thermal transients [3,4]. Therefore, TES systems give CSP plants an edge over photovoltaics or wind power [5]. Solid sensible heat storage using concrete as the storage material is expected to be an attractive option due to its lower investment and maintenance costs [6].

Concrete storage is a regenerative storage system where the storage module is cyclically heated by hot and cooled by cold heat transfer fluid (HTF) flows. A tubular heat exchanger with a defined tube pitch is typically imbedded in the storage concrete for the HTF flow [7]. The German Aerospace Center (DLR) successfully tested a concrete storage module on the Plataforma Solar de Almeria in Spain in 2003 [6]. After that, a 20 m^3 solid media storage test module connected to an electrically heated thermal oil loop was built in Stuttgart for live tests and further improvements [5,7,8]. Skinner et al. [9] and John et al. [10] developed a high performance concrete which was tested in a concrete storage unit at 450 °C using molten nitrate salt as the HTF. Combined storage systems using concrete as the sensible storage module were also tested by Bai et al. [11] and Laing et al. [12,13].

The heat transfer between the HTF and the solid material is an important issue which is reviewed in this work. Zhang and Faghri [14,15] presented a semi-analytical method to analyze the heat transfer in a hollow cylindrical storage unit with a phase change material (PCM) as the storage material. Bai et al. [16] also used a semi-analytical method to analyze the entire heat transfer process

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Nomenclature

a	inner radius of the cylindrical energy storage unit [m]
b	outer radius of the cylindrical energy storage unit [m]
Bi	Biot number
Bi_E	corrected Biot number
C	specific heat [J/kg K]
d_h	tube wall thickness [m]
d_i	inner diameter of the energy storage unit [m]
d_o	outer diameter of the energy storage unit [m]
$f(p)$	Laplace transform of T_{in}
h	heat transfer coefficient [W/(m^2 K)]
h_E	corrected heat transfer coefficient [W/(m^2 K)]
k	thermal conductivity [W/m K]
L	energy storage unit length [m]
$\dot{m}$	mass flow rate [kg/s]
m_s	solid mass of the storage unit [kg]
M	equilibrium temperature [K]
N	number of heat transfer tubes
p	Laplace transform parameter
P	heat transfer surface perimeter of the cylindrical surface [m]
Q	delivered energy [kJ]
Q_{dis}	discharging energy [kWh]
$r_{concrete}$	concrete cost per unit mass [$/kg]
r_{steel}	steel cost per unit mass [$/kg]
$R_{concrete}$	concrete cost of storage module [$]
R_{steel}	steel cost of storage module [$]
S_f	wetted area in the tube [m^2]
S_h	heat transfer area [m^2]
S_s	section area of solid material [m^2]
t	time [s]
t_{dis}	discharge time [h]
T	temperature [°C]
$T_{f,accepted}$	accepted fluid temperature [°C]
$T_{f,out}$	fluid outlet temperature [°C]
T_0	storage unit initial temperature [°C]
T_∞	fluid inlet temperature [°C]
$T_{s,mean}$	mean temperature of storage unit [°C]
ΔT	inlet and outlet temperature difference [K]
$<t>$	weighted average time [s]
U	average fluid velocity in the axial direction in the cylindrical energy storage unit [m/s]
V_s	solid storage module volume [m^3]
z	longitudinal direction [m]

Greek symbols

α_s	thermal diffusivity of the solid storage material ($=k_s/(\rho_s C_s)$) [m^2/s]
β	intermediate parameters in Eq. (22) ($=1+\beta_f/\beta_r$)
β_f	coefficient in Eq. (16) ($=P/(\rho_f S_f C_f)$)
β_s	coefficient in Eq. (17) ($=P/(\rho_s S_s C_s)$)
ρ	density [kg/m^3]
τ	total time for the charging process [s]
τ_t	thermal time constant ($=\rho_s C_s V_s/(hS_h)$)
Λ	Laplace transform of ΔT
φ	energy utilization of the solid storage module

Subscripts

f	heat transfer fluid
s	solid storage material

Superscripts

*	dimensionless variable
a	value for $r = a$

Acronyms

CSP	concentrating solar power
HTF	heat transfer fluid
PCM	phase change material
TES	thermal energy storage

inside a hollow cylinder in a solid media. A non-steady model was developed by Salomoni et al. [17] to analyze the heat transfer in a hollow cylindrical concrete storage unit. However, all of these methods used iterative method to solve the equations, which can be very time consuming.

To simplify the analysis of the heat transfer between the HTF and the solid storage material, Bradshaw et al. [18] extended the use of the lumped capacitance method to conditions where the internal thermal resistance of the solid is significant compared to the external convective thermal resistance. Since most solid storage materials have very low thermal conductivities, the method will not be accurate if the assumption for the lumped heat capacitance heat conduction method in a solid storage material is not satisfied. Thus, they introduced an effective heat transfer coefficient for spherical particles as a correction to the heat transfer coefficient. In their work, the convective heat transfer coefficient, h, was replaced by an effective heat transfer coefficient, h_E, in the governing equations. This effective heat transfer coefficient was used with the lumped capacitance method to analyze the heat storage/extraction from the solid spherical particles. Xu et al. [19] used the methodology developed by Bradshaw to develop formulas for the effective heat transfer coefficient for three types of thermal storage structures, plates, cylindrical rods immersed in the fluid and a solid core with circular channels inside for the HTF. Effective heat transfer coefficients for four typical solid storage designs were also summarized by Schmidt et al. [20]. Wu et al. [21] investigated the impact of the concrete structure on the thermal performance of a tank-type concrete TES system using the effective heat transfer coefficient developed by Xu et al. [19]. However, although the fluid temperature variations in the charging/discharging processes for the solid storage system were significant, both Bradshaw et al. [18] and Xu et al. [19] neglected the transient terms in the heat transfer fluid in the governing equations when deriving the effective heat transfer coefficient.

The present work derives an effective heat transfer coefficient for a cylindrical heat storage unit using the methodology developed by Bradshaw et al. [18] with the analysis including the transient terms in the energy equation for the HTF. Then, the model is validated with experimental data for a concrete storage module. Finally, a new design method that considers both the thermal storage and the material costs is developed for a solid storage module using the modified lumped capacitance method.

2. Thermal energy storage model

Fig. 1 shows a typical solid medium sensible heat storage module with a tubular heat exchanger with collectors and distributors for the HTF embedded in the solid storage material.

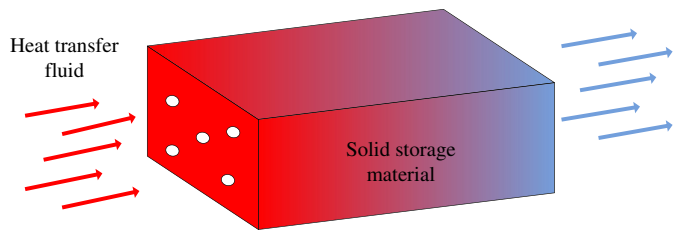

Fig. 1. Solid medium sensible heat storage module.

A cross section of the solid storage module is shown in Fig. 2. The tubes embedded in the storage module were arranged in a triangular pitch to obtain a better temperature distribution in the solid storage module. The entire storage module consisted of several storage units with regular hexagonal cross sections. The analysis of the storage unit was then simplified by substituting cylindrical storage unit geometry for the regular hexagonal shapes with the same cross-sectional area. The geometries were related by $d_o^2 = 2\sqrt{3}D^2/\pi$.

The model is based on the following assumptions: the storage module has the same thermal behavior as the cylindrical heat storage unit shown in Fig. 3; thermal conduction in the axial direction in the fluid is negligible; axial heat conduction in the solid can be neglected; the HTF directly contacts the solid and the tube thickness can be assumed to be zero; there are no heat losses from the storage unit to the surroundings; and the lumped capacitance method is applicable to the thermal conduction in the solid media in the radial direction; thus, $\partial T_s/\partial r = 0$. The 2D transient model for the solid storage unit can then be simplified to a 1D model.

Assuming that the fluid mass flow rate, $\dot{m}$, in the tubes is constant, the thermal energy balance for the fluid in the differential control volume, dz, in Fig. 3 is [22]:

$$h_E P \mathrm{d}z\left(T_s - T_f\right) + \dot{m}C_f T_{f,z} = \dot{m}C_f T_{f,z+\mathrm{d}z} + \rho_f S_f \mathrm{d}z C_f \frac{\partial T_f}{\partial t} \tag{1}$$

where h_E is effective heat transfer coefficient in the tube, P is the heat transfer surface perimeter ($P = \pi \cdot d_i$), and S_f is the wetted area in the tube ($S_f = (\pi \cdot d_i^2)/4$).

Substituting the average fluid velocity in the cylindrical heat storage unit ($U = \dot{m}/(\rho_f S_f)$), and rearranging Eq. (1), the fluid energy balance equation becomes:

$$\frac{\partial T_f}{\partial t} + U\frac{\partial T_f}{\partial z} = \frac{h_E P}{\rho_f S_f C_f}\left(T_s - T_f\right) \tag{2}$$

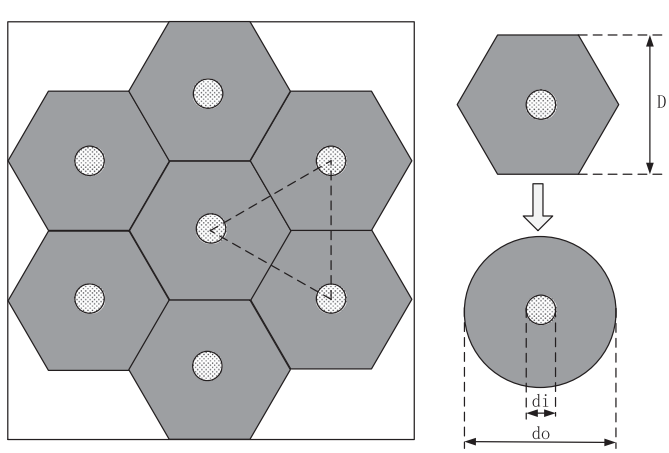

Fig. 2. Cross section of the solid storage module.

The energy balance for the solid storage material uses the same control volume, dz, as shown in Fig. 3. The solid storage material energy balance is:

$$h_E P \mathrm{d}z\left(T_s - T_f\right) = -\rho_s S_s \mathrm{d}z C_s \frac{\partial T_s}{\partial t} \tag{3}$$

where S_s is the sectional area of the solid material ($S_s = \pi \cdot (d_o^2 - d_i^2)/4$). Rearranging Eq. (3), the energy balance equation can be written as:

$$\frac{\partial T_s}{\partial t} = -\frac{h_E P}{\rho_s S_s C_s}\left(T_s - T_f\right) \tag{4}$$

Van Lew et al. [23] developed a very efficient numerical solution for this model using the method of characteristics. This numerical solution method can be used to quickly solve the energy balance equations, Eqs. (2) and (4).

The next section derives the effective heat transfer coefficient for the cylindrical heat storage unit using a methodology similar to that developed by Bradshaw et al. [18] with the transient terms retained in the fluid energy equation.

3. Derivation of the effective heat transfer coefficient

The weighted average time, which was introduced by Bradshaw et al. [18], is defined as:

$$<t> = \frac{\int_0^\tau t \cdot \Delta T(t)\mathrm{d}t}{\int_0^\tau \Delta T(t)\mathrm{d}t} \tag{5}$$

where $\Delta T(t)$ is the difference between the inlet and outlet fluid temperatures, τ(τ tends to infinity) is the charging time. The Laplace transform of the temperature difference and its derivative are:

$$L\left[\Delta T(t)\right] = \int_0^\tau \Delta T(t) e^{-pt}\mathrm{d}t = \Lambda(p) \tag{6}$$

$$\frac{\mathrm{d}\Lambda(p)}{\mathrm{d}p} = -\int_0^\tau t\Delta T(t) e^{-pt}\mathrm{d}t = \Lambda'(p) \tag{7}$$

Therefore, the weighted average time can be calculated as:

$$<t> = \frac{\int_0^\tau t \cdot \Delta T(t)\mathrm{d}t}{\int_0^\tau \Delta T(t)\mathrm{d}t} = -\frac{\Lambda'(0)}{\Lambda(0)} \tag{8}$$

This weighted average time refers to the charging time when the storage unit is charged to some percent of fully charged, which can be used as a characteristic parameter of the storage unit for the charging process. The weighted average time for the charging of the storage unit should be the same for any method used to analyze this process.

The following sections consider two cases where the lumped capacitance method is applicable to the heat conduction in the solid storage unit and then when the temperature gradient in the radial direction must be considered. Analytical formulas are obtained for the weighted average times for both two cases and compared to determine the effective heat transfer coefficient, h_E.

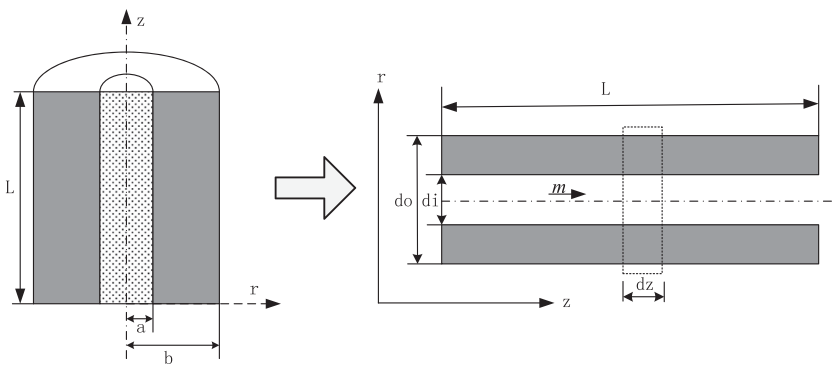

Fig. 3. Cylindrical heat storage unit model.

3.1. Weighted average time for the lumped capacitance method

The coupled energy equations for the fluid and the solid of the heat storage unit shown in Fig. 3 are Eqs. (2) and (4). The initial and boundary conditions are:

$$t = 0, \quad T_f = T_s = 0 \tag{9}$$

$$t = \tau(\tau \to \infty), \quad T_f = T_s = M \tag{10}$$

$$z = 0, \quad T_f = T_{\text{in}}(t) \tag{11}$$

where T_{in} is the fluid inlet temperature. Further assume that the solid material reaches the final equilibrium temperature, M, at the end of the charging process at time τ(τ tends to infinity).

The Laplace transforms of the governing equations are

$$Me^{-p\tau} + p\Lambda_f + U\frac{\partial \Lambda_f}{\partial z} = h_E\beta_f(\Lambda_s - \Lambda_f) \tag{12}$$

$$Me^{-p\tau} + p\Lambda_s = -h_E\beta_s(\Lambda_s - \Lambda_f) \tag{13}$$

where $\beta_f = P/(\rho_f S_f C_f)$, $\beta_s = P/(\rho_s S_s C_s)$.

The solution of Eqs. (12) and (13) gives:

$$\Lambda_f = f(p)e^{-p\sigma z} + \left(e^{-p\sigma z} - 1\right)\frac{Me^{-p\tau}}{p} \tag{14}$$

where: σ=$(1 + h_E\beta_f/(p + h_E\beta_s))/U$ and$f(p) = \int_0^\tau T_{\text{in}}e^{-pt}\mathrm{d}t = L[T_{\text{in}}]$. Then, the Laplace transform of the temperature difference is:

$$L\left[\Delta T(t)\right] = f(p)\left(1 - e^{-p\sigma L}\right) + \frac{Me^{-p\tau}}{p}\left(1 - e^{-p\sigma L}\right) = \Lambda_1(p) \tag{15}$$

Thus, the weighted average time is:

$$\langle t\rangle = -\frac{\Lambda_1'(0)}{\Lambda_1(0)} = \frac{\beta_f}{h_E\beta_s^2\beta} + \frac{L}{2U}\beta + \tau - \frac{f(0)}{M} \tag{16}$$

where $\beta = 1 + \beta_f/\beta_r$.

3.2. Weighted average time considering the temperature gradient in the radial direction

The coupled energy equations for the fluid and the solid storage unit are:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T_s}{\partial r}\right) = \frac{1}{\alpha_s}\frac{\partial T_s}{\partial t} \tag{17}$$

$$\frac{\partial T_f}{\partial t} + U\frac{\partial T_f}{\partial z} = h\beta_f\left(T_s^a - T_f\right) \tag{18}$$

where: $\alpha_s = k_s/(\rho_s C_s)$. The initial and boundary conditions are:

$$r = a, \quad -k_r\frac{\partial T_s}{\partial r} = h\left(T_f - T_s\right) \tag{19}$$

$$r = b, \quad \frac{\partial T_s}{\partial r} = 0 \tag{20}$$

$$t = 0, \quad T_f = T_s = 0 \tag{21}$$

$$t = \tau, \quad T_f = T_s = M \tag{22}$$

$$z = 0, \quad T_f = T_{\text{in}}(t) \tag{23}$$

The Laplace transforms of Eqs. (17)–(20) are

$$\frac{\partial^2 \Lambda_s}{\partial r^2} + \frac{1}{r}\frac{\partial \Lambda_s}{\partial r} - \frac{p}{\alpha_s}\Lambda_s = \frac{M}{\alpha_s}e^{-p\tau} \tag{24}$$

$$\left(Me^{-p\tau} + p\Lambda_f\right) + U\frac{\partial \Lambda_f}{\partial z} = h\beta_f\left(\Lambda_s^a - \Lambda_f\right) \tag{25}$$

$$k_s\frac{\partial \Lambda_s}{\partial r} = h\left(\Lambda_s - \Lambda_f\right), \quad r = a \tag{26}$$

$$\frac{\partial \Lambda_s}{\partial r} = 0, \quad r = b \tag{27}$$

The solution of Eqs. (24) and (25) gives:

$$\Lambda_f = \left[f(p) + \frac{Me^{-p\tau}}{p}\right]\exp\left(-\frac{p+\varepsilon}{U}z\right) - \frac{Me^{-p\tau}}{p} \tag{28}$$

where $\varepsilon = g\beta_{ab}\cdot h\beta_f/(g\beta_{ab} - h/k_s\cdot\beta_{0b})$, $\beta_{0b} = I_{0a}K_b + I_bK_{0a}$, $\beta_{ab} = I_aK_b - I_bK_a$, g =$(p/\alpha_s)^{0.5}$, $I_{0a} = I_0(ag)$, $K_{0a} = K_0(ag)$, $I_a = I_1(ag)$,

$I_b = I_1(bg)$, $K_a = K_1(ag)$, $K_b = K_1(bg)$ and I_0, I_1, K_0, K_1 are the first and second kinds of the modified Bessel functions. Then, the Laplace transform of the temperature difference is:

$$L\left[\Delta T(t)\right] = f(p)\left[1 - \exp\left(-\frac{p+\varepsilon}{U}L\right)\right] + \frac{Me^{-p\tau}}{p}\left[1 - \exp\left(-\frac{p+\varepsilon}{U}L\right)\right] = \Lambda_2(p) \tag{29}$$

As p tends to zero, the asymptotic formulas for the modified Bessel functions [24] are $K_0(x) = \ln(2/x)$ and $K_1(x) = 1/x$. Thus, the weighted average time is:

$$\langle t\rangle = -\frac{\Lambda_2'(0)}{\Lambda_2(0)} = -\frac{\lim\limits_{p\to 0}\frac{\varepsilon''}{2}}{\beta} + \frac{L}{2U}\beta + \tau - \frac{f(0)}{M} \tag{30}$$

where $\lim\limits_{p\to 0}\varepsilon'' = (h\beta_f/(2(\alpha_s h/k_s)^2))\cdot[h/k_s\cdot((a^4 - 4a^2b^2 + 3b^4)/4a - (b^4/a)\ln(b/a)) - ((a^2-b^2)^2/a^2)]$.

Finally, comparing Eqs. (16) and (30), the effective heat transfer coefficient for the cylindrical heat storage unit is:

$$\frac{1}{h_E} = \frac{1}{h} + \frac{1}{k_s}\cdot\frac{4ab^4\ln\frac{b}{a} - 3ab^4 + 4a^3b^2 - a^5}{4(b^2-a^2)^2} \tag{31}$$

where a (=$d_i/2$) and b (=$d_o/2$) are the inner and outer radii of the cylindrical storage unit.

The relationship between the convective coefficient, h, and effective convective coefficient, h_E, in Eq. (31) is the same as the relationship in Xu et al. [19], but the effective heat transfer coefficient in Eq. (31) is derived with the transient terms retained in the fluid energy equation, since the fluid temperature varies during the charging/discharging processes for the solid energy storage system. Therefore, the validity of the effective heat transfer coefficient in Eq. (31) can be extended to the condition of the fluid temperature varying during the charging/discharging processes.

4. Model validation

4.1. Validation of the effective heat transfer coefficient

This section describes the results of the energy storage predictions using the lumped capacitance method (with h), modified lumped capacitance method (with h_E) and an analytical method.

The general solution for the transient heat conduction using the lumped capacitance method can be easily found in a heat transfer book. The energy delivered from the solid material is [22]:

$$Q = \rho_s C_s V_s\left(T_0 - T_\infty\right)\left[1 - \exp\left(-\frac{t}{\tau_t}\right)\right] \tag{32}$$

where T_0 is the initial temperature of the solid material, T_∞ is the fluid temperature, V_s is the solid volume ($V_s = S_s\cdot L$), τ_t is the thermal time constant ($\tau_t = \rho_s C_s V_s/(hS_h)$) and S_h is the heat transfer area ($S_h = \pi d_i\cdot L$).

The following dimensionless variables are used to simplify:

$$Q^* = \frac{Q}{\rho_s C_s V_s(T_0 - T_\infty)}; \quad t^* = \frac{k_s}{\rho_s C_s}\cdot\frac{t}{(V_s/S_h)^2} \tag{33}$$

Then, the dimensionless energy delivered from the solid for the lumped capacitance method and the modified lumped capacitance method are:

$$Q^*(t^*) = 1 - \exp(-Bi\cdot t^*) \tag{34}$$

$$Q_E^*(t^*) = 1 - \exp(-Bi_E\cdot t^*) \tag{35}$$

where Bi is the Biot number for the cylindrical heat storage unit (Fig. 3), which is:

$$Bi = \frac{h}{k_s}\cdot\frac{V_s}{S_h} = \frac{h}{k_s}\cdot\frac{b^2-a^2}{2a} \tag{36}$$

$$Bi_E = \frac{h_E}{k_s}\cdot\frac{b^2-a^2}{2a} = \frac{Bi}{1 + Bi\cdot\frac{4\eta^4\ln(\eta) - 3\eta^4 + 4\eta^2 - 1}{2(\eta^2-1)^3}} \tag{37}$$

where η is the diameter ratio of the storage unit ($\eta = b/a$).

Ozisik [25] gives the analytical solution for the transient heat conduction from a hollow cylinder in a fluid by using the separation of variables method. The solid temperature is:

$$\theta(r,t) = \sum_{m=1}^{\infty}\frac{T_0 - T_\infty}{N(\beta_m)}e^{-\alpha_s\beta_m^2 t}R_0(\beta_m, r)\int_a^b r'R_0(\beta_m, r')\mathrm{d}r' \tag{38}$$

$$\frac{1}{N(\beta_m)} = \frac{\pi^2}{2}\frac{\beta_m^2\left[\beta_m J_1(\beta_m a) + \frac{h}{k_s}J_0(\beta_m a)\right]^2}{\left[\beta_m J_1(\beta_m a) + \frac{h}{k_s}J_0(\beta_m a)\right]^2 - \left[\left(\frac{h}{k_s}\right)^2 + \beta_m^2\right]\cdot J_1^2(\beta_m b)} \tag{39}$$

$$R_0(\beta_m, r) = J_1(\beta_m b)Y_0(\beta_m r) - Y_1(\beta_m b)J_0(\beta_m r) \tag{40}$$

where $\theta(r,t) = T_s - T_\infty$, $J_v(\beta_m r)$ and $Y_v(\beta_m r)$ are the Bessel function of order v of the first and second kind. The eigenvalues, β_m, are the positive roots of the transcendental equation:

$$\beta_m\left[Y_1(\beta_m b)J_1(\beta_m a) - Y_1(\beta_m a)J_1(\beta_m b)\right] = \frac{h}{k_s}\left[J_1(\beta_m b)Y_0(\beta_m a) - Y_1(\beta_m b)J_0(\beta_m a)\right] \tag{41}$$

Using the definition of the dimensionless variables in Eq. (33), the dimensionless energy delivered from the solid is:

$$Q_{AN}^* = 1 - \frac{2}{\eta^2-1}\sum_{m=1}^{\infty}\tilde{N}\left(\overline{\beta}_m\right)\cdot e^{-\left(\frac{\eta^2-1}{2}\right)^2\cdot\overline{\beta}_m^2 t^*}\cdot\left[J_1(\overline{\beta}_m)Y_1(\eta\overline{\beta}_m) - J_1(\eta\overline{\beta}_m)Y_1(\overline{\beta}_m)\right]^2 \tag{42}$$

$$\tilde{N}\left(\overline{\beta}_m\right) = \frac{\pi^2}{2}\cdot\frac{\left[\overline{\beta}_m J_1(\overline{\beta}_m) + \overline{Bi}J_0(\overline{\beta}_m)\right]^2}{\left[\overline{\beta}_m J_1(\overline{\beta}_m) + \overline{Bi}J_0(\overline{\beta}_m)\right]^2 - \left[\overline{Bi}^2 + \overline{\beta}_m^2\right]J_1^2\left(\eta\overline{\beta}_m\right)} \tag{43}$$

where $\overline{Bi} = a\cdot h/k_s = 2Bi/(\eta^2-1)$ and the eigenvalues $\overline{\beta}_m(= a\beta_m)$ are the positive roots of:

$$\overline{\beta}_m\left[Y_1(\eta\overline{\beta}_m)J_1(\overline{\beta}_m) - Y_1(\overline{\beta}_m)J_1(\eta\overline{\beta}_m)\right] = \overline{Bi}\left[J_1(\eta\overline{\beta}_m)Y_0(\overline{\beta}_m) - Y_1(\eta\overline{\beta}_m)J_0(\overline{\beta}_m)\right] \quad (44)$$

The dimensionless delivered energy predicted by the lumped capacitance method (Eq. (34)), the modified lumped capacitance method (Eq. (35)) and the analytical method (Eq. (42)) are shown in Fig. 4(a)–(c) for various *Bi* for three diameter ratios.

Fig. 4 shows that the dimensionless energy predictions of the modified lumped capacitance method are in very good agreement with the analytical method for Biot numbers from 0.1 to 100 for all three diameter ratios. The results from all three methods are the same when the Biot number is 0.1. However, when *Bi* is bigger than 1, the results from the lumped capacitance method are very different with these for both the modified lumped capacitance method and the analytical method. The results from the modified lumped capacitance method agree well with the results of the analytical method for *Bi* equal to 10. Even for *Bi* equal to 100, the difference between the modified lumped capacitance method and the analytical method are very small. With increasing diameter ratio, η, from 4 to 12, the difference between the modified lumped capacitance method and the analytical method become even smaller. For a fixed Biot number, the effective Biot number in Eq. (37) will become smaller as the diameter ratio increases. Thus, the modified lumped capacitance method will be more accurate at large diameter ratio.

The modified lumped capacitance method using the effective heat transfer coefficient, h_E, can provide sufficient accuracy for solid energy storage models. The analytical method assumes that the solid temperature in the hollow cylindrical storage unit is uniform and that the fluid temperature does not vary with time or position in the heat transfer tube during the charging process. The modified lumped capacitance method does not have any limitations on the initial temperature of the solid material or the fluid temperature. Thus, the modified lumped capacitance method is more general than the analytical method introduced by Ozisik [25].

4.2. Comparison of the numerical results with experimental data

The thermal response of a concrete storage module was tested to evaluate the modeling work. The experiments used water as the heat transfer fluid and concrete as the storage material. The concrete had a density of 3100 kg/m^3, a specific heat of 1.18 kJ/(kg K) and a thermal conductivity of 2.65 W/m K [26].

The concrete storage module contained a tube bank consisting of 86 tubes with inner diameters of 16 mm. The tubes were arranged in a triangular pitch with a pitch distance of 100 mm as shown in Fig. 5(a). The module had two 4 m long concrete storage blocks. The second block was placed on the top of the first block with the two storage blocks assembled in series as shown in Fig. 5(b). Pictures of the concrete storage module and tube bank are shown in Fig. 6.

The simulation results were compared with the experimental data with the same hot water inlet temperature and volumetric flow rate as in the experiment. The storage unit in the model was initialized to the temperature measured in the experiment at steady state before charging. Fig. 7 compares the predicted and measured axial temperature profile in the storage unit for various volume flow rates. The average solid temperature is difficult to estimate, there are some differences between the simulations and experimental results but with the same tendencies.

Fig. 8 compares the numerical and experimental results for different volume flow rates. The energy collected by the solar collector is different because the experiments for these four cases were tested on four different days. Thus, the fluid inlet temperatures for these four cases have a little difference. The numerical results for the outlet fluid temperature agree well with the experimental data. Some differences between the numerical and experimental results can be attributed to measurement errors and heat losses from the module which were not considered in the model. Therefore, the model gives a reasonable description of the thermal behavior of the solid storage unit.

5. Results and discussion

5.1. Storage module optimization goal and system requirements

The optimization goal is to minimize the material cost (including the costs of the storage material and the heat transfer tube material) per unit storage capacity. The optimization constraints are that the storage module must achieve the system requirements. The fluid outlet temperature and the energy released from storage module are two key indexes of the storage system for solar thermal power plant applications.

The first requirement of the storage system is that during the discharging process, the fluid outlet temperature from the storage system cannot be lower than the lowest fluid temperature accepted by the power block. Since the fluid outlet temperature decreases continually during the discharging process, the first requirement means that the fluid outlet temperature must be equal to the lowest allowable fluid temperature at the end of the discharge, which is $T_{f,\mathrm{out}}\big|_{t=t_{\mathrm{dis}}} = T_{f,accepted}$.

The second requirement of the storage system is that the storage module must release the energy Q_{dis} during the discharge time t_{dis}. The energy utilization for the solid storage unit is defined based on the temperature of the solid storage material as:

$$\varphi = \frac{Q_{\mathrm{delivery}}}{Q_{\mathrm{storage}}} = \frac{m_s C_s (T_0 - T_{s,\mathrm{mean}})}{m_s C_s (T_0 - T_\infty)} \quad (45)$$

where $T_{s,\mathrm{mean}}$ is the mean temperature of the solid storage material and m_s is the solid mass of the storage unit ($m_s = \rho_s S_s L$).

Then, the released energy from the storage module during the discharge time is:

$$Q_{\mathrm{dis}} = N \cdot m_s C_s (T_0 - T_\infty) \cdot \varphi|_{t=t_{\mathrm{dis}}} \quad (46)$$

where N is the number of heat transfer tubes in the storage module, which is also the number of storage units.

Define the costs per unit mass (kg) of the heat transfer tube and storage material as $r_{\mathrm{steel}}$and r_{concrete}. The costs of the heat transfer tube material, R_{steel}, and the storage material, R_{concrete}, are then:

$$R_{\mathrm{steel}} = LN \cdot \frac{\pi}{4}\left[(d_i + 2d_h)^2 - d_i^2\right] \cdot \rho_{\mathrm{steel}} \cdot r_{\mathrm{steel}} = NL\pi\left(d_h^2 + d_i d_h\right)\rho_{\mathrm{steel}} r_{\mathrm{steel}} \quad (47)$$

$$R_{\mathrm{concrete}} = m_s r_{\mathrm{concrete}} \cdot N = N \rho_s S_s L r_{\mathrm{concrete}} \quad (48)$$

where d_i is inner tube diameter, d_h is the tube wall thickness and L is the storage unit length.

The material cost per the unit storage capacity is then:

$$r_{\mathrm{material}} = \frac{R_{\mathrm{material}}}{Q_{\mathrm{dis}}} = \frac{R_{\mathrm{steel}} + R_{\mathrm{concrete}}}{N \cdot m_s C_s (T_0 - T_\infty) \cdot \varphi|_{t=t_{\mathrm{dis}}}} = \frac{\pi\left(d_h^2 + d_i d_h\right)\rho_{\mathrm{steel}} r_{\mathrm{steel}} + \rho_s S_s r_{\mathrm{concrete}}}{\rho_s S_s C_s (T_0 - T_\infty) \cdot \varphi|_{t=t_{\mathrm{dis}}}} \quad (49)$$

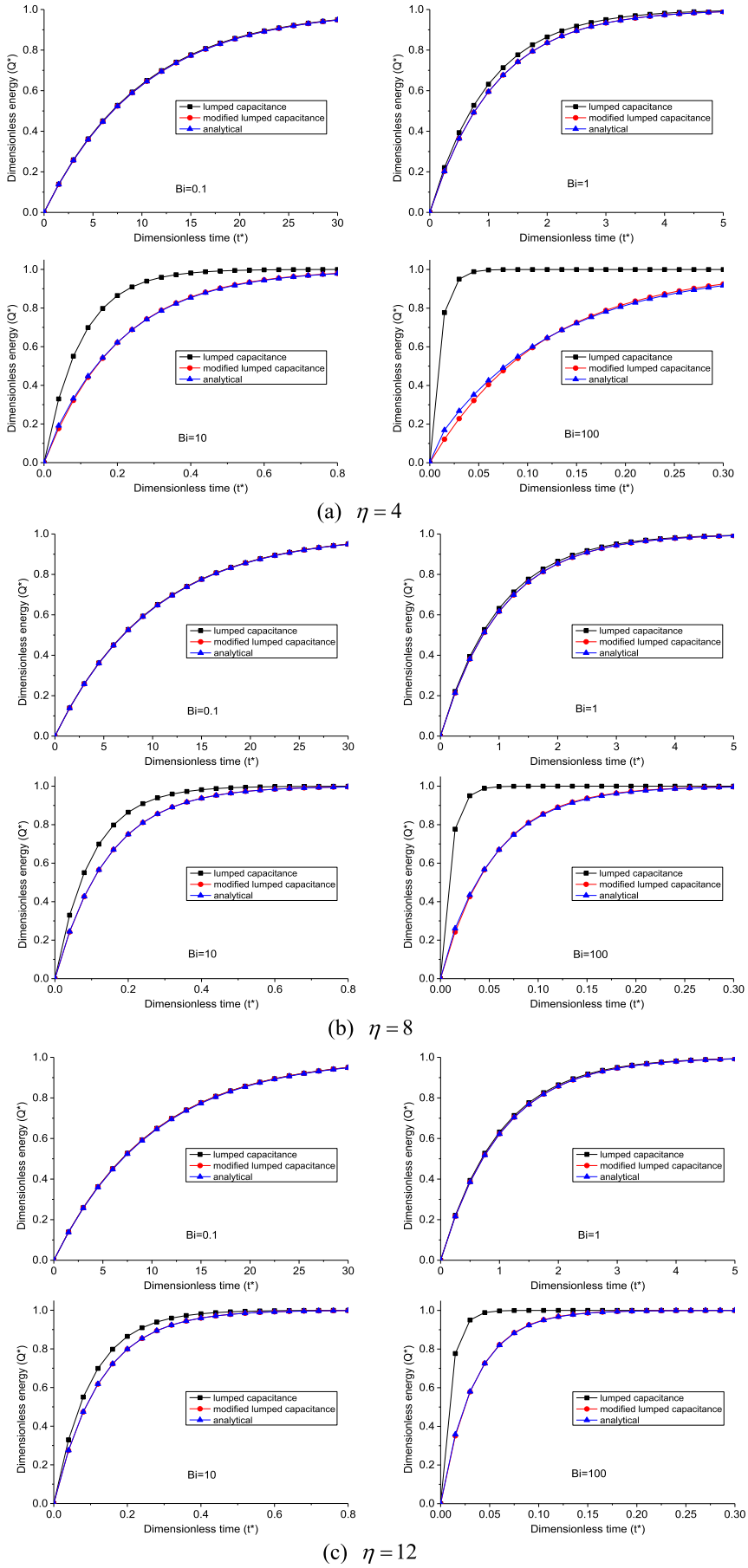

Fig. 4. Predictions of the lumped capacitance method, modified lumped capacitance method and analytical method for various *Bi* for three diameter ratios.

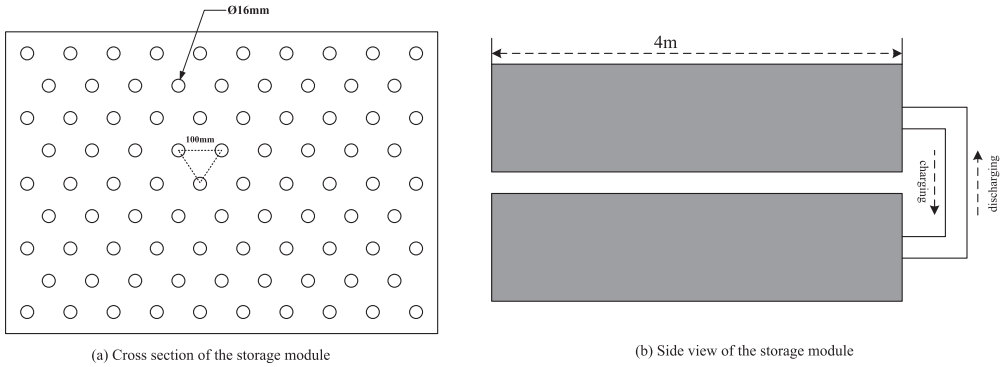

Fig. 5. The cross section and side view of the storage module.

Fig. 6. Concrete storage module and tube bank: (a) before insulating, (b) after insulating.

5.2. Solid storage module design procedure

The solid storage module design includes selection of the diameter ratio, η, storage length, L, average fluid velocity, U, number of heat exchanger tubes, N, inner tube diameter, d_i, and tube wall thickness, d_h.

Since there are national standards for choosing the inner diameter, d_i, and the wall thickness, d_h, of the heat transfer tube,

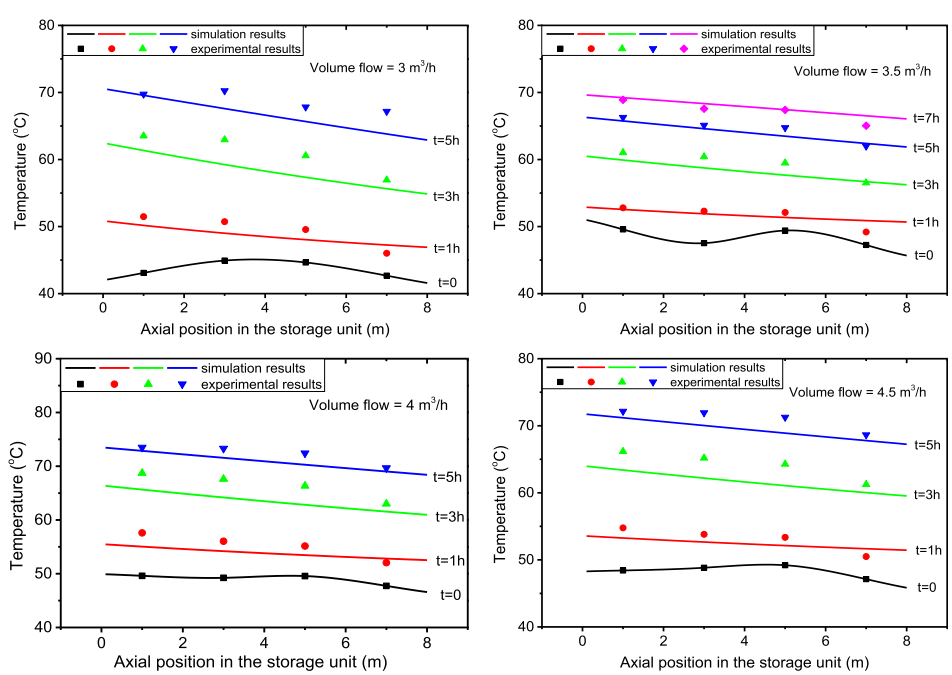

Fig. 7. Predicted and measured axial temperature profile in the storage unit.

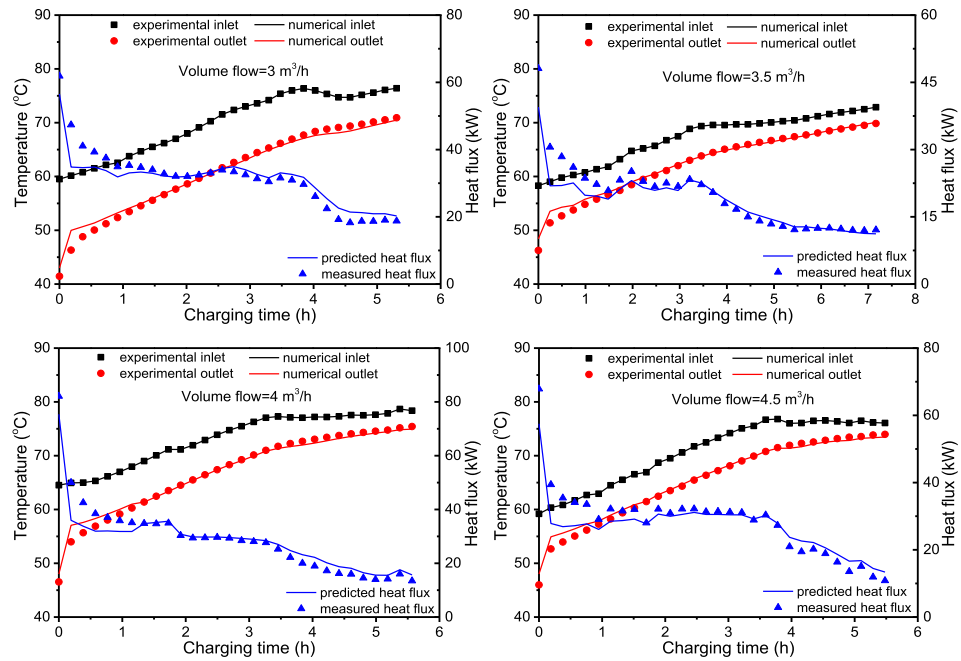

Fig. 8. Comparison of numerical and experimental results for different volume flow rates.

the inner diameter and thickness of the tube were selected from d_i = 16 mm, 18 mm or 20 mm and d_h = 1 mm, 2 mm or 3 mm according to the pressure requirements of the heat transfer fluid. According to Eq. (49), the number of heat transfer tubes, N, does not affect the material cost per unit storage capacity, but only the total energy released from the storage module. The designer can select the another three parameters, the diameter ratio, storage length and average fluid velocity, according to the required fluid outlet temperature to minimize the material costs and then decide the number of heat transfer tubes based on the released energy.

The analysis leads to the following design procedure for the solid storage module:

(a) Find possible combinations of the diameter ratio, η, storage length, L, and average fluid velocity, U, using the modified lumped capacitance method that give the required fluid temperature, $T_{f,\text{accepted}}$, for the discharge time, t_{dis}. The details for this step are:
  i) Select the minimums, maximums and step sizes of the diameter ratio ($\eta_{\min}$, $\eta_{\max}$, $d\eta$) and the storage length ($L_{\min}$, $L_{\max}$, dL).
  ii) Fix the diameter ratio and the storage length at $\eta_i(=\eta_{\min} + i \cdot d\eta, i = 0, 1, \ldots, m)$ and $L_j\,(=L_{\min} + j \cdot dL, j = 0, 1, \ldots, n)$.
  iii) Change the average fluid velocity to make the fluid outlet temperature equal to the required fluid temperature at time t_{dis}.
  iv) Record these design parameter combinations (η_i, L_i, U_i, $i = 1, 2, \ldots, (m+1) \times (n+1)$).
(b) Calculate the energy utilization of the storage module, φ, in Eq. (45) using the design parameter combinations.
(c) Calculate the material cost per unit storage capacity in Eq. (49), compare the calculate results and find the optimized design parameters, η_{opt}, L_{opt} and U_{opt}, to minimize the material cost per unit storage capacity, r_{material}.
(d) Use the required energy released from the storage module, Q_{dis}, to determine the number of heat transfer tubes, N_{opt}, using Eq. (46).

5.3. Preliminary optimized the solid storage module design

Concrete was used as the solid storage material with thermal oil VP-1 as the heat transfer fluid in this design. The thermo physical properties of the concrete and thermal oil are listed in Table 1.

The costs per unit mass (kg) of the heat transfer tube and storage material are $r_{\text{steel}} = 2.2$ \$/kg [28] and $r_{\text{concrete}} = 0.05$ \$/kg [29]. The initial storage unit temperature is $T_0 = 390\,^\circ$C, the fluid inlet temperature is $T_\infty = 290\,^\circ$C, and the steel density is $\rho_{\text{steel}} = 7850$ kg/m^3. The storage system requirement and the ranges of some parameters are listed in Table 2.

The acceptable combinations of the diameter ratio, η, storage length, L, and average fluid velocity, U, for the fluid temperature and discharge time constraints are shown in Fig. 9 for all the parameter ranges. The storage module fluid outlet temperatures for the parameter combinations in Fig. 9 all satisfy fluid temperature constraint at the end of the discharge.

Table 1
Thermo physical properties of the concrete and thermal oil [5,27].

Material	Density (kg/m^3)	Specific heat (J/kg K)	Thermal conductivity (W/m K)	Viscosity (Pa s)
Concrete (400 °C)	2250	1050	1.20	–
Therminol VP-1 (360 °C)	749	2480	0.0844	0.00017

Table 2
The system requirement and the selection range of some parameters.

Parameters	Value
Released energy (MWh)	10
Discharging time (h)	6
Accepted fluid temperature (°C)	350
Diameter ratio η	6–10
Storage length (m)	30–200
Inner diameter (mm)	16
Thickness of the tube (mm)	1

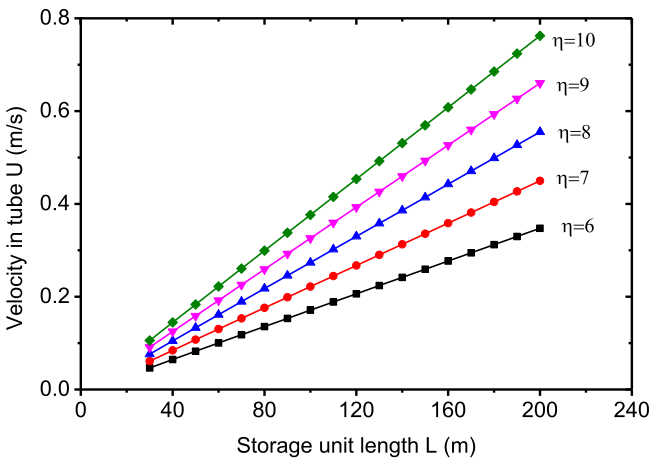

Fig. 9. Acceptable design parameter combinations.

The storage module energy utilization calculated by the modified lumped capacitance method for all acceptable combinations of the design parameters are shown in Fig. 10.

For the same diameter ratio, the storage module energy utilization increases with the storage length. Smaller diameter ratio gives better energy utilization for a fixed storage length. According to Eq. (49), the material cost per unit storage capacity, $r_{material}$, is only affected by the energy utilization and the diameter ratio with an inverse relation between the energy utilization and the material cost. Thus, for the same diameter ratio, a longer storage length will give a higher energy utilization, which will lower the material cost per unit storage capacity. Therefore, the diameter ratio should be determined firstly and then the storage length.

The material costs per unit storage capacity for all allowable combinations of the design parameters are shown in Fig. 11.

The material costs are very similar for diameter ratios of 8 and 9 and much lower than the others. For storage lengths longer than 150 m, the material cost for a diameter ratio equal to 8 is lower than for a diameter ratio equal to 9. Therefore, for a given heat flux (released energy and discharge time in Table 2), $\eta_{opt} = 8$ is the optimized diameter ratio for this design. The material cost per unit storage capacity is nearly constant for storage lengths longer than 160 m but longer storage lengths lead to more power for the circulating pump. Therefore, a length of $L_{opt} = 160$ m was selected as the optimized storage length.

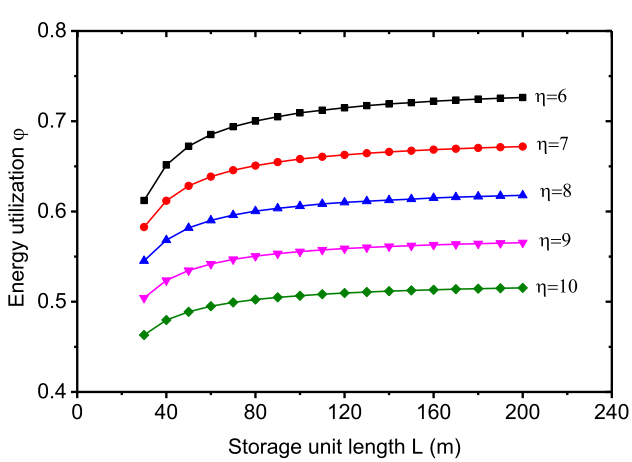

Fig. 10. Storage module energy utilization for all allowable combinations of the design parameters.

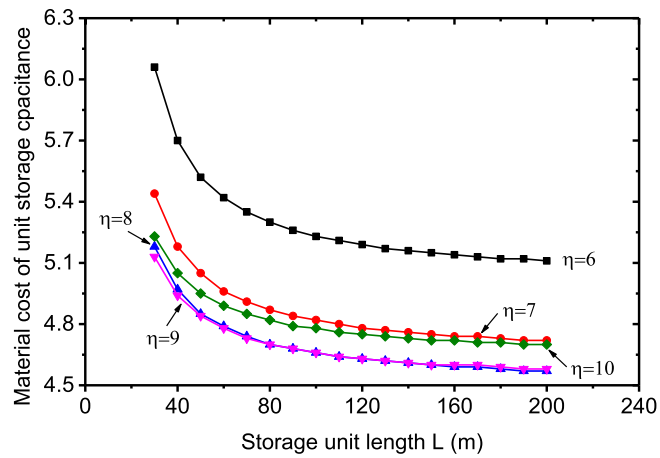

Fig. 11. Material costs per unit storage capacity for the all allowable design combinations.

Table 3
Preliminary optimized design parameters.

Design parameter	Diameter ratio η_{opt}	Storage length L_{opt}	Velocity in tube U_{opt}	Number of heat transfer tubes N_{opt}
Results	8	160 m	0.4426 m/s	123

Using Eq. (48), the number of heat transfer tubes would then be:

$$N_{opt} = \frac{Q_{dis}}{m_s C_s (T_0 - T_\infty) \cdot \varphi|_{t=t_{dis}}} = 122.2 \approx 123 \tag{50}$$

The optimized design parameter results for the system requirements in Table 2 are then collected in Table 3.

6. Conclusions

The modified lumped capacitance method was used with an effective heat transfer coefficient in a simplified analysis of the heat transfer in solid thermal energy storage systems. An initial model was developed using the modified lumped capacitance method for a solid cylindrical heat storage unit. The effective heat transfer coefficient in the model is defined with consideration of the transient terms in the fluid energy equation. The model is compared against both analytical results and experimental data for validation. The experiments used water as the heat transfer fluid and concrete as the storage material. The tests showed that the current model can be used to accurately describe the thermal behavior of solid storage units during charging and discharging. A design procedure is given to optimize solid storage module designs with a design optimization case given as an example.

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